YUE TONG

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London, United Kingdom

Profile

PhD-trained translational scientist with five years developing iPSC-derived organoid disease models, CRISPR genome editing, and integrative molecular analyses at ULB. Experienced in multi-institution international collaboration, bridging patient genetics, experimental biology, and transcriptomics to drive mechanistic understanding. Since September 2026, applying this scientific background as Business Development Manager (Product & Services) at Biocytogen, alongside continued part-time cancer evolution research at the University of Surrey.

Skills

Cell & Molecular Biology: CRISPR/Cas genome editing; isogenic cell line generation; human iPSC culture and differentiation; organoid systems; cell-based functional assays; proliferation and viability assays; transfection; RNA interference; flow cytometry; ELISA; Western blotting; RT-qPCR; Seahorse metabolic assays
Quantitative & Computational Biology: tumour phylogenetics; likelihood-based inference; transcriptomics; multi-omics integration; reproducible workflow development
Programming: R; Python; Bash; LaTeX
Imaging & Data Analysis: CellProfiler; ImageJ/Fiji; quantitative microscopy; GraphPad Prism; SPSS
Clinical & Translational: physician licensure (China); clinical cohort design; biomarker analysis; phenotype stratification
Languages: English (professional proficiency); Mandarin (native); Cantonese (conversational); French (conversational)

Research Experience

Business Development Manager — Product & Services Sep. 2026 – Present
Biocytogen *London, GB*

- **Scientific business development for Biocytogen's preclinical technology portfolio**
  Support business development for Biocytogen's preclinical technology portfolio, spanning humanized and customized mouse models (BioMice™) and preclinical pharmacology services. Advise prospective clients on technical scope and study design, support scientific evaluation ahead of contract discussions, and represent Biocytogen's scientific capabilities at conferences and trade shows.

  Keywords: business development; preclinical models; BioMice; preclinical pharmacology; scientific communication

Visiting Researcher (Part-time) Nov. 2025 – Present
School of Biosciences, University of Surrey *Guildford, UK*

- **Cancer evolution and copy-number–driven tumour phylogenetics**
  Develop reproducible computational workflows for large-scale biological datasets and benchmark statistical models for evolutionary inference. Collaborate within an interdisciplinary research environment to support data-driven biological discovery.

  Keywords: reproducible workflows; quantitative analysis; biological datasets; statistical modelling

Doctoral Fellow Sep. 2020 – Sep. 2025
ULB Center for Diabetes Research, Université Libre de Bruxelles *Brussels, Belgium*
Fund For Research Training in Industry & Agriculture Fellow, F.R.S.-FNRS *Brussels, Belgium*

- **Mechanistic modeling of rare and common diabetes using gene editing and organoid technologies**
  Conceived, led and independently executed multi-year research programs establishing 8 gene-edited, isogenic iPSC-derived pancreatic beta-cell models integrating clinical phenotypes with multi-omics datasets. Dissected variant-specific pathogenic mechanisms, elucidating links between genetic perturbations and *insulin biosynthesis defects*, *ER–Golgi stress responses*, and *gluco-lipotoxicity*. Delivered end-to-end experimental strategy from hypothesis formulation to functional validation and quantitative data interpretation.

Keywords: iPSC; organoids; CRISPR/Cas; isogenic systems; patient-derived samples; monogenic and polygenic diseases; functional genomics

- **Nutrient metabolism and developmental regulation of pancreatic beta cells: cross-species analysis**
  Led the human experimental arm of a cross-species project investigating beta-cell development. Combined genome editing, transcriptomics, and functional assays across human and mouse models to study mechanisms regulating cellular differentiation and function.

  Keywords: cross-species models; nutrient metabolism; developmental biology; transcriptomics; functional assays

Graduate Research Fellow Sep. 2017 – Jun. 2020
The Second Xiangya Hospital, Central South University *Hunan, Changsha, China*

- **Serum proinsulin dynamics across diabetes subtypes**
  Designed and executed a clinical translational study involving 300 participants to characterize subtype-specific proinsulin secretion dynamics. Established standardized recruitment and measurement protocols and performed statistical analyses to identify biomarker signatures distinguishing diabetes phenotypes.

  Keywords: clinical study design; biomarker discovery; phenotype stratification; statistical analysis

- **Immunologic and genetic architecture of autoimmune diabetes**
  Coordinated large-scale cohort enrollment and longitudinal follow-up of autoimmune diabetes patients and first-degree relatives. Conceptualized and led a disease prediction and staging framework integrating neutrophil RNA sequencing and GWAS data to characterize immunogenetic risk architecture and early disease trajectories.

  Keywords: cohort study; immunogenetics; RNA sequencing; GWAS; disease prediction; translational research

Education

ULB Center for Diabetes Research, Université Libre de Bruxelles Brussels, Belgium
PhD of Biomedical and Pharmaceutical Science *Sep. 2020 – Sep. 2025*

The Second Xiangya Hospital, Central South University Hunan, Changsha, China
Master of Clinical Medicine: Internal Medicine *Sep. 2017 – Jun. 2020*

Xiangya School of Medicine, Central South University Hunan, Changsha, China
Bachelor of Clinical Medicine *Sep. 2012 – Jun. 2017*

Publications

1. **Tong Y**, Becker M, Schierloh U et al. A new form of diabetes caused by INS mutations defined by zygosity, stem cell and population data. EMBO Mol Med (2026). https://doi.org/10.1038/s44321-025-00362-9
2. Lytrivi M*, **Tong Y***, Virgilio E* et al. Diabetes mellitus and the key role of endoplasmic reticulum stress in pancreatic beta cells. Nat Rev Endocrinol. 2025 Jun 4. https://doi.org/10.1038/s41574-025-01129-5. Co-first author.
3. Van Mulders A, Willems L, Coenen S et al., **Tong Y** et al. A critical role for iron import through the transferrin receptor in developing beta-cells. Nat Commun 17, 2826 (2026). https://doi.org/10.1038/s41467-026-69574-y
4. Wu R, Álvarez-Carretero S, **Tong Y** et al. Integrated analysis of fossils and molecular divergence time estimates a latest Jurassic origin of angiosperms. Nat Plants 12, 1242–1251 (2026). https://doi.org/10.1038/s41477-026-02311-x
5. Mandla R, Lorenz K, Yin X et al., **Tong Y** et al. Multi-omics characterization of type 2 diabetes associated genetic variation. medRxiv [Preprint]. 2024 Jul 15:2024.07.15.24310282
6. Bourgeois S, Van Mulders A, Heremans Y et al., **Tong Y** et al. ER stress relief drives beta-cell proliferation. (Accepted in Diabetologia)
7. Piron A, Szymczak F, Folon L et al., **Tong Y** et al. Identification of novel type 1 and type 2 diabetes genes by co-localization of human islet eQTL and GWAS variants with colocRedRibbon. Cell Genomics. 2025 Sep 16:101004.
8. Arunagiri A, Alam M, Haataja L et al., **Tong Y** et al. Proinsulin folding and trafficking defects trigger a common pathological disturbance of endoplasmic reticulum homeostasis. Protein Science. 2024;33(4):e4949.
9. **Tong Y**, Yang L, Shao F et al. Distinct secretion pattern of serum proinsulin in different types of diabetes. Ann Transl Med. 2020;8(7):452.
10. Xing Y, Lin Q, **Tong Y** et al. Abnormal Neutrophil Transcriptional Signature May Predict Newly Diagnosed Latent Autoimmune Diabetes in Adults of South China. Front Endocrinol (Lausanne). 2020;11:581902.
11. Hu J, Liu Z, **Tong Y** et al. Fibroblast Growth Factor 19 Levels Predict Subclinical Atherosclerosis in Men With Type 2 Diabetes. Front Endocrinol (Lausanne). 2020;11:282.

Awards and Scholarships

ISPAD Standard Travel Grant | *International Society for Pediatric and Adolescent Diabetes (ISPAD)* Oct. 2023

SFD Travel Grant for Research Meetings in Diabetes | *French-Speaking Diabetes Society (SFD)* Oct. 2023

Fund For Research Training in Industry & Agriculture Doctoral Fellowship | *F.R.S.-FNRS* Oct. 2021

First-Prize Academic Scholarship | *Central South University* Sep. 2017

Third-Prize School Scholarship | *Central South University* Sep. 2013

Conference Talks and Posters

2024–2025 Homozygous and heterozygous *INS* mutations cause divergent clinical and iPSC-derived beta-cell phenotypes, *The International Group on Insulin Secretion meeting of EASD*, Schloss Hohenkammer, Germany & *The 9th Meeting of Study Group on Genetics of Diabetes (SGGD)*, Exeter, UK | Talks

2023 Discovery of a new treatment for a novel form of rare diabetes caused by an insulin gene mutation using patients' iPSC-derived beta-cells, *The European Association for the Study of Diabetes (EASD) 2024 Annual Meeting & The International Society for Pediatric and Adolescent Diabetes (ISPAD) 2024 Annual Meeting*, Rotterdam, the Netherlands | Talks

2023 Bedside-inspired diabetes modeling: learn from monogenic diabetes to understand the pathogenic mechanisms of T1D, *19th Immunology of Diabetes Society (IDS) Congress*, Paris, France | Invited talk

2020 Distinct secretion pattern of serum proinsulin in different types of diabetes, *15th Xiangya International Diabetes Immunology Forum & 17th Immunology of Diabetes Society (IDS) Congress*, Beijing, China | Poster

References

Available upon request. Primary referees: Dr Bingxin Lu (Lecturer, University of Surrey); Dr Ruolin Wu (Research Fellow, University of Surrey); Prof Miriam Cnop (Professor, ULB); Prof Decio Eizirik (Professor, ULB).